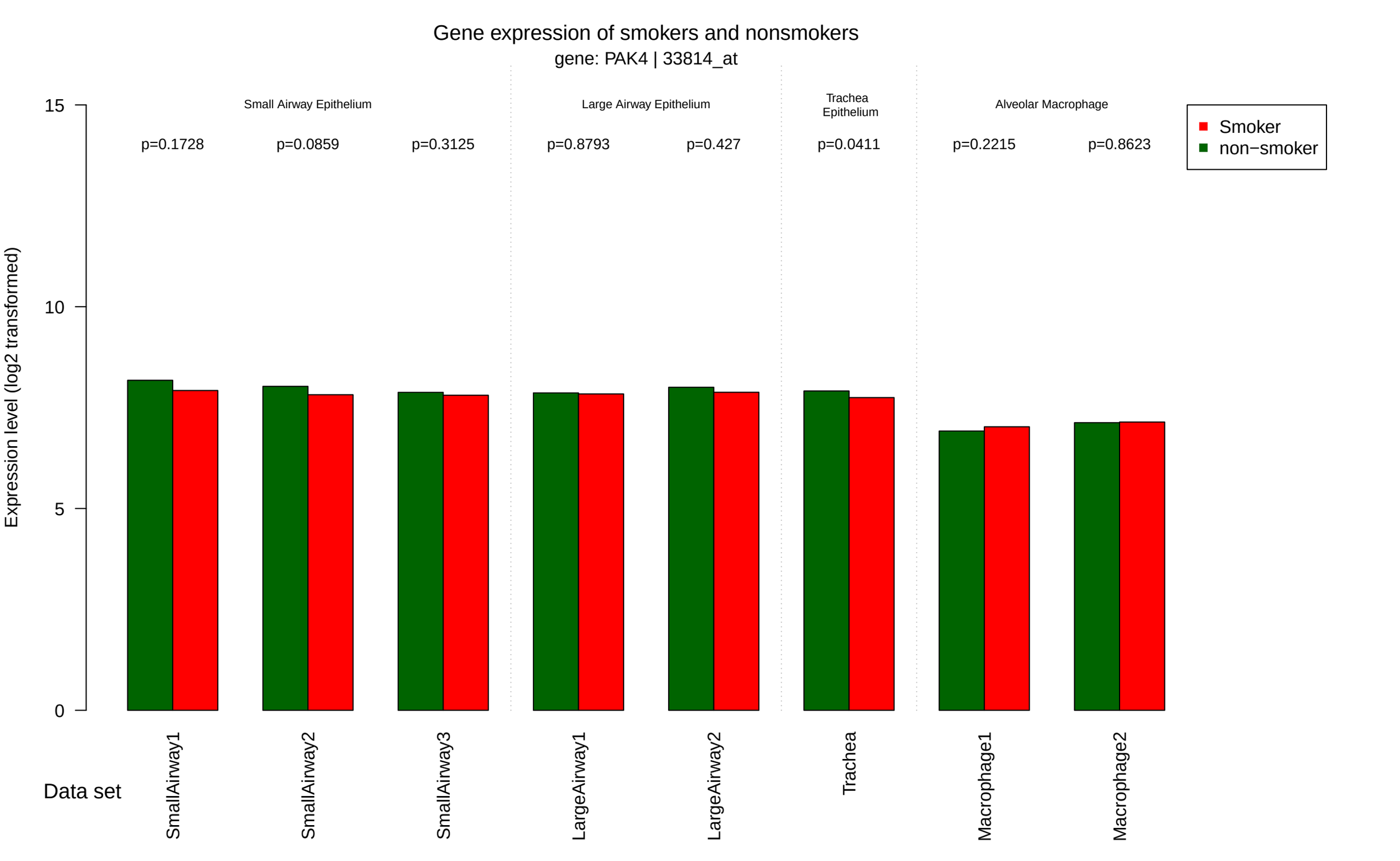
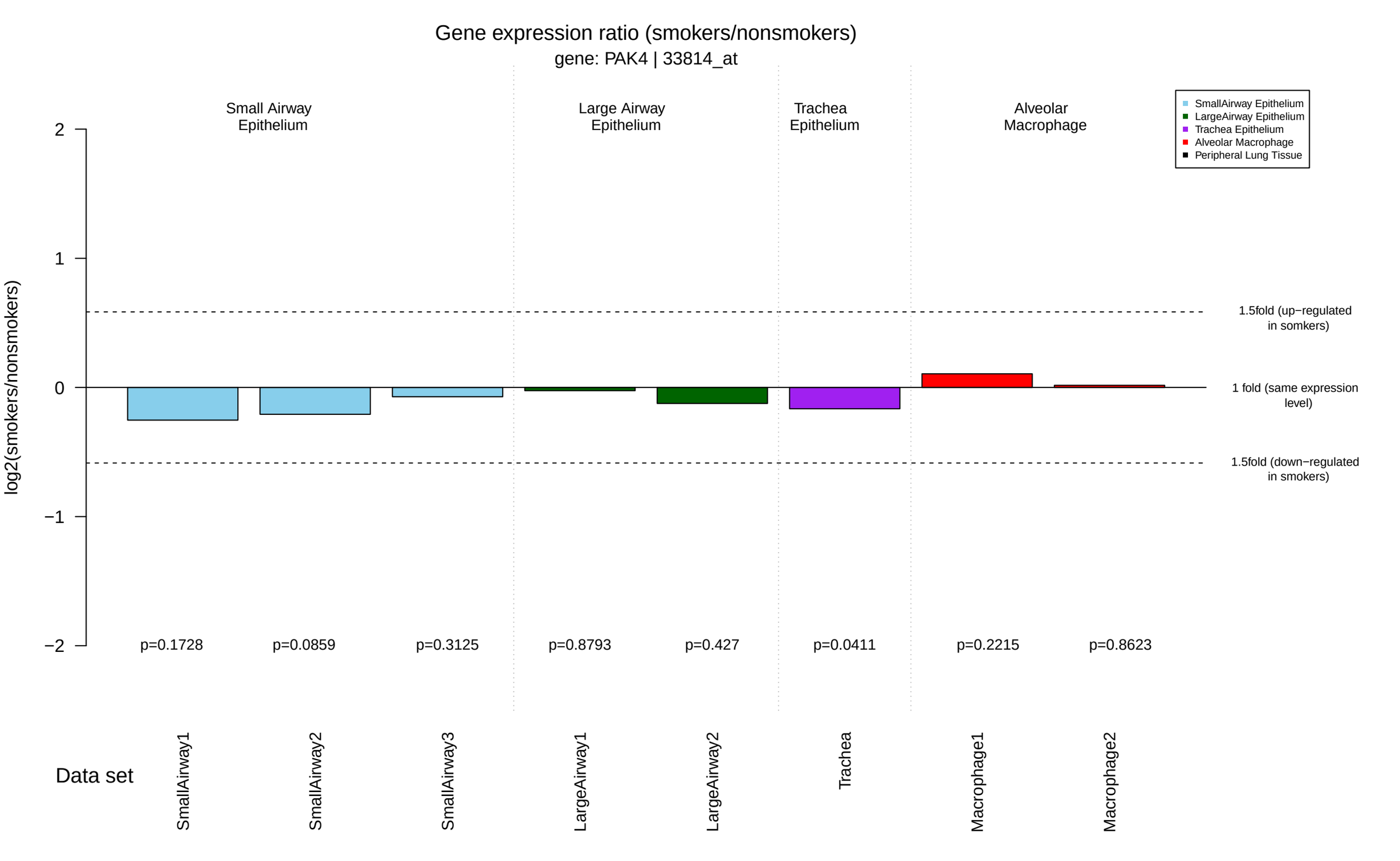
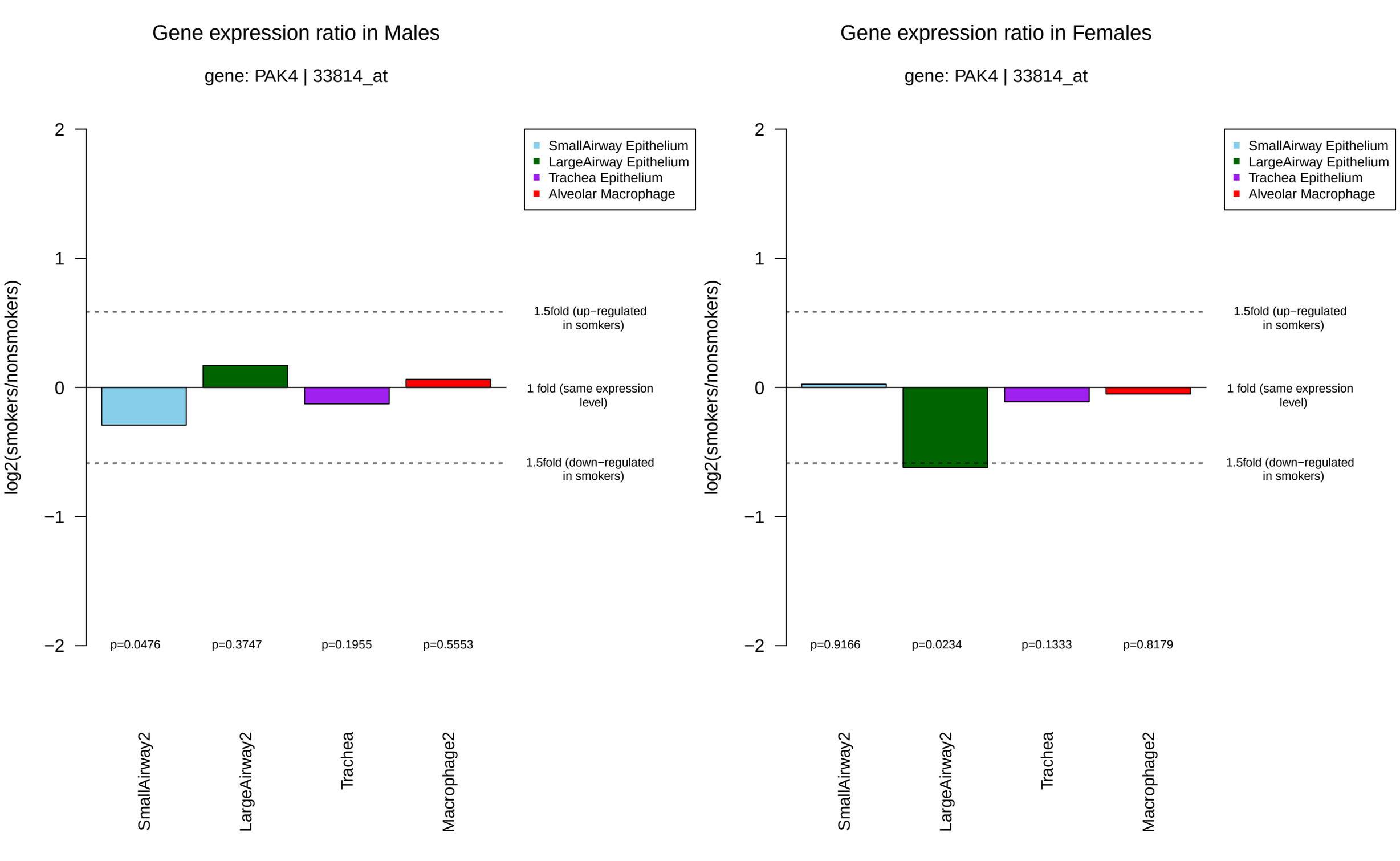
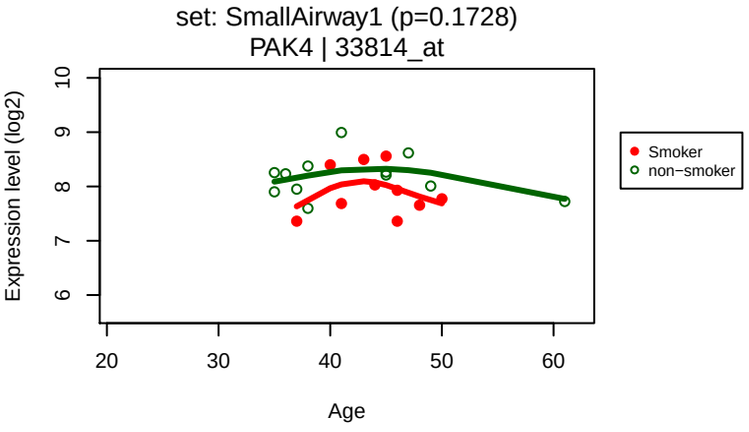
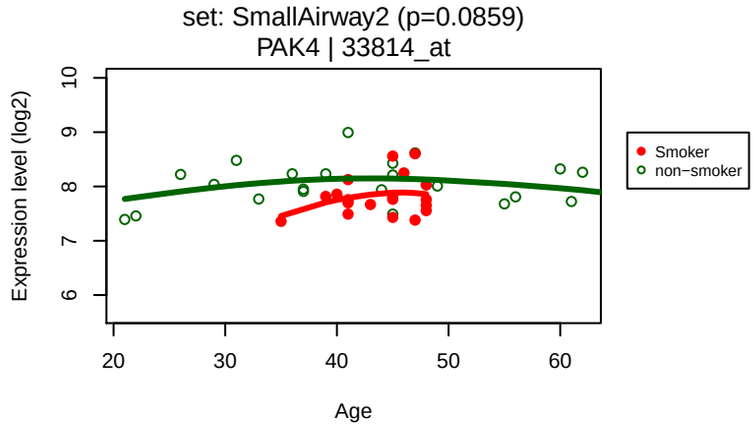
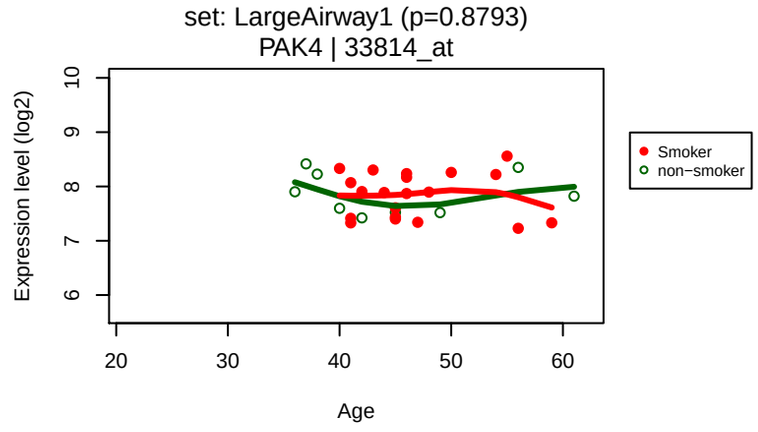

Gene expression of smokers and nonsmokers
gene: PAK4 | 33814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1728
p=0.0859
p=0.3125
p=0.8793
p=0.427
p=0.0411
p=0.2215
p=0.8623
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

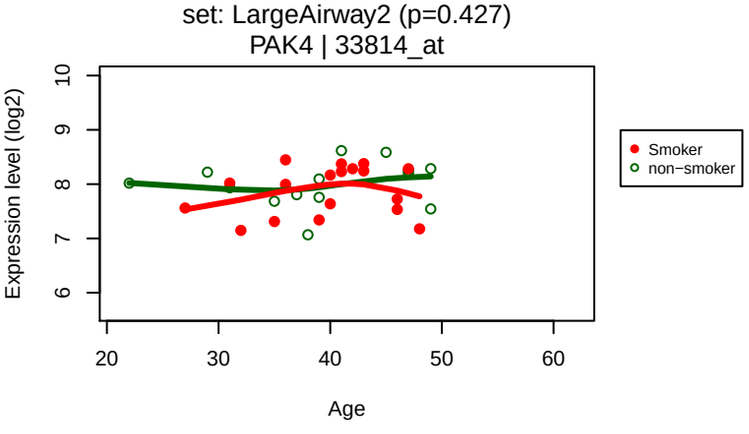

Gene expression ratio (smokers/nonsmokers)
gene: PAK4 | 33814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1728
p=0.0859
p=0.3125
p=0.8793
p=0.427
p=0.0411
p=0.2215
p=0.8623
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

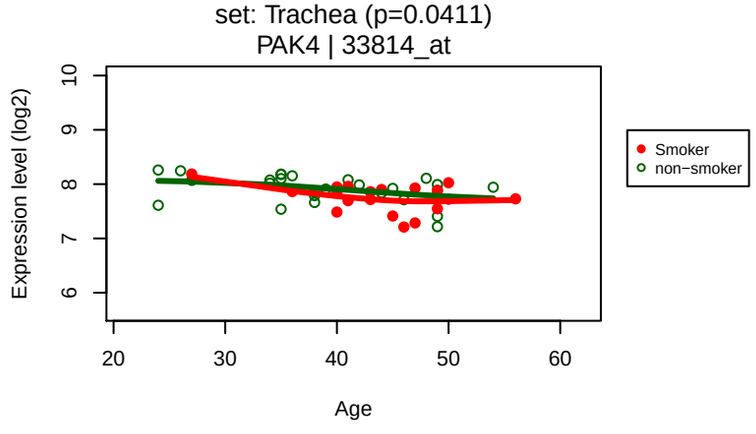

Gene expression ratio in Males
gene: PAK4 | 33814_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0476
p=0.3747
p=0.1955
p=0.5553
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK4 | 33814_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9166
p=0.0234
p=0.1333
p=0.8179
SmallAirway2
LargeAirway2
Trachea
Macrophage2

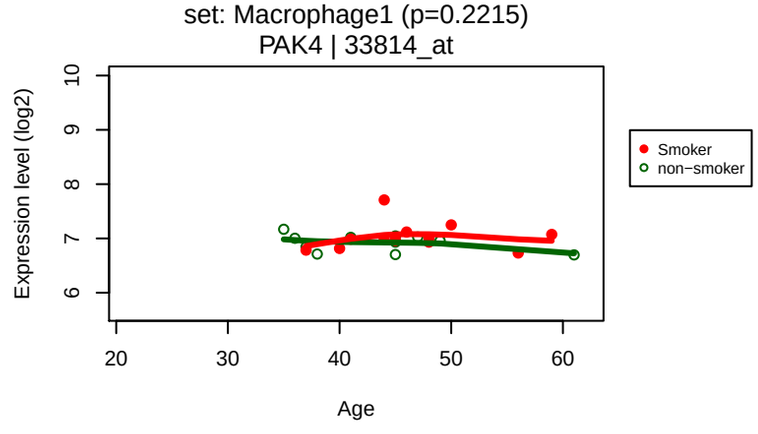

set: SmallAirway1 (p=0.1728)
PAK4 | 33814_at
Expression level (log2)
Age
Smoker
non−smoker

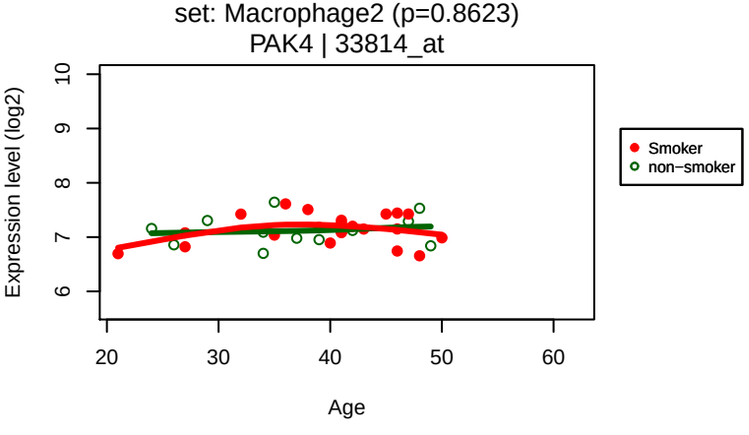

set: SmallAirway2 (p=0.0859)
PAK4 | 33814_at
Expression level (log2)
Age
Smoker
non−smoker

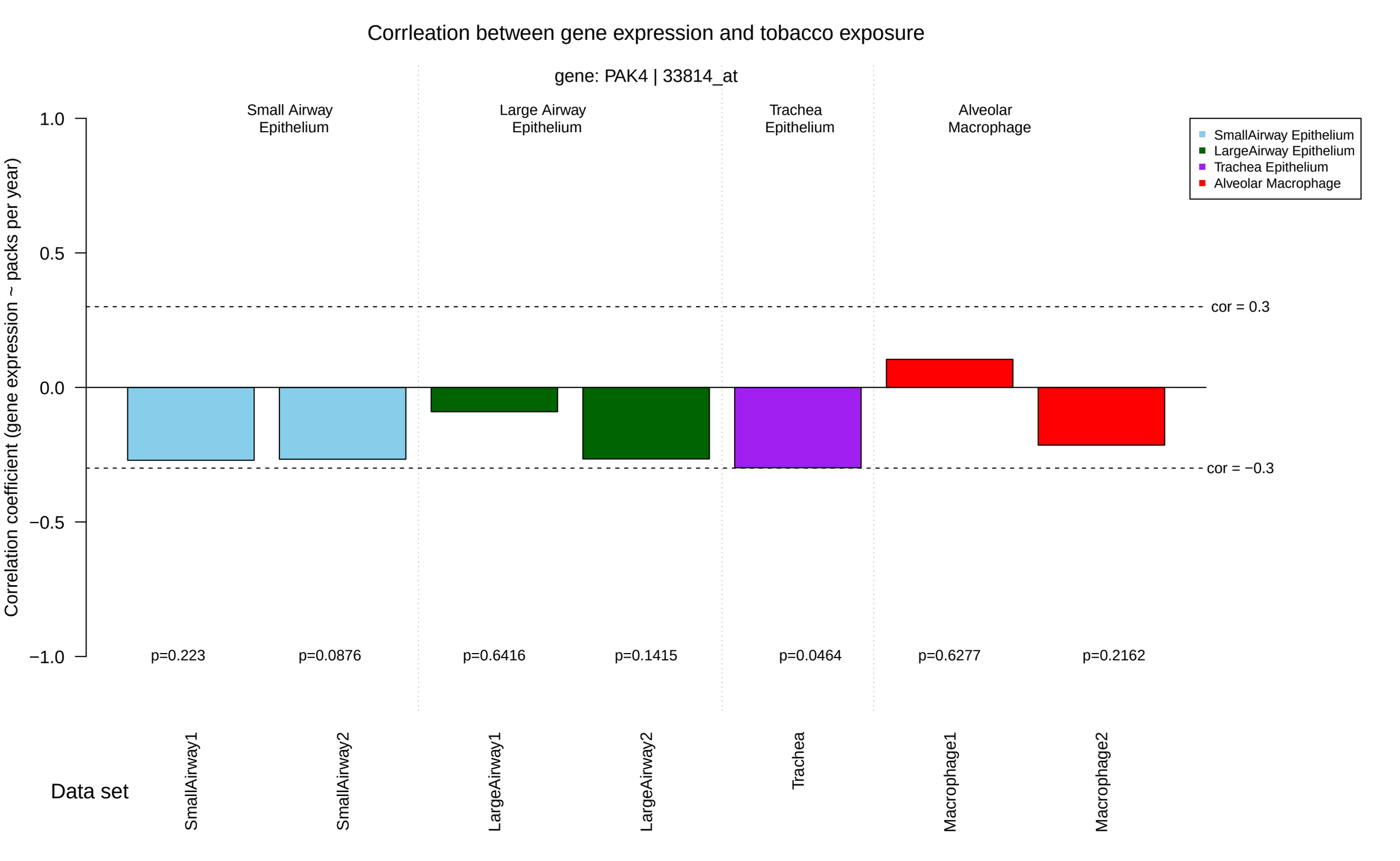

set: LargeAirway1 (p=0.8793)
PAK4 | 33814_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.427)
PAK4 | 33814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0411)
PAK4 | 33814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2215)
PAK4 | 33814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8623)
PAK4 | 33814_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK4 | 33814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.223
p=0.0876
p=0.6416
p=0.1415
p=0.0464
p=0.6277
p=0.2162
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2